NANYANG TECHNOLOGICAL UNIVERSITY SINGAPORE

IE4727 Web Application Design

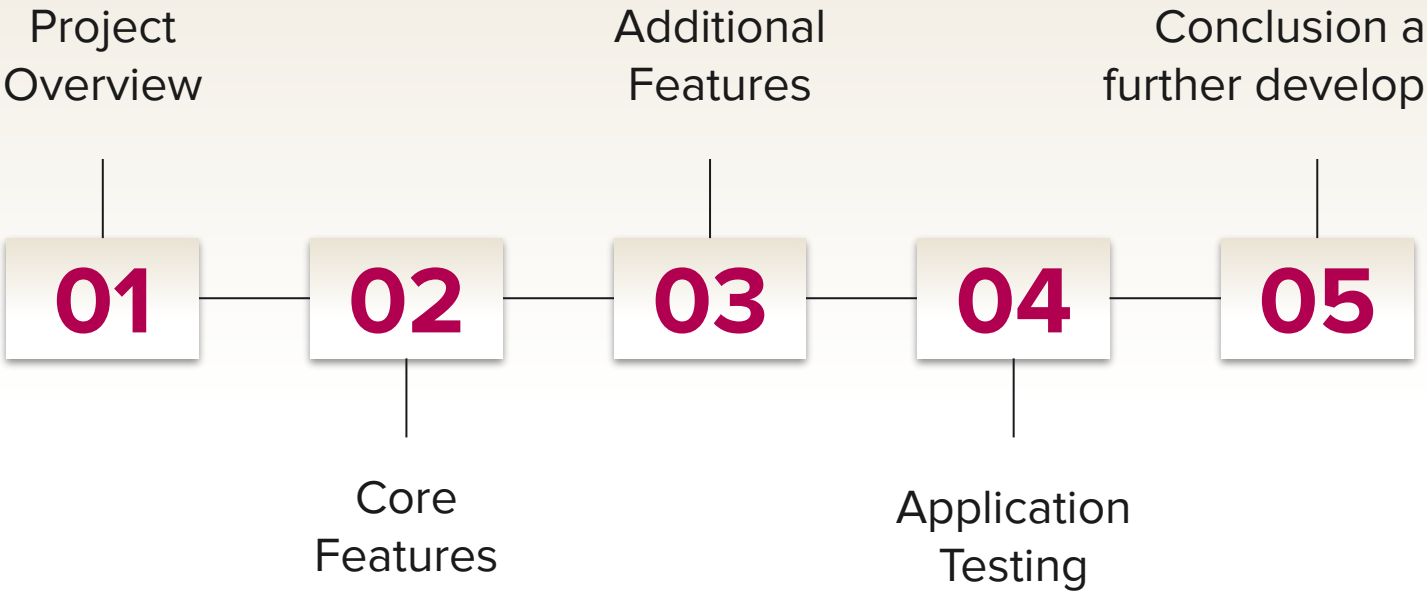
Design Project

Instructor: Dr Hu Xiao



Student name: Nguyen Tuan Anh & Nguyen Minh Duc
Group F32-DG07

Presentation Flow



1. Project overview

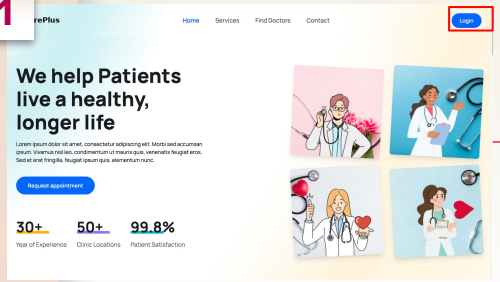
CarePlus is an all-in-one healthcare website aimed to enhance the interaction between patients and doctors. The platform provides a seamless user experience for browsing doctor profiles and scheduling appointments based on their availability. It furthermore allows flexibility to reschedule appointments for both doctors and patients and sends a confirmation email upon successful booking.

Besides the basic features, CarePlus also includes a built-in video call feature between doctors and patients, and a chatbot capable of answering common inquiries, providing pre-diagnosis information, and recommendations of matched doctors based on the pre-diagnosis. With the application of Generative AI and a more flexible means of communication, CarePlus not only facilitates better virtual consultation, but also ensures a seamless, efficient, and patient-friendly approach to healthcare.

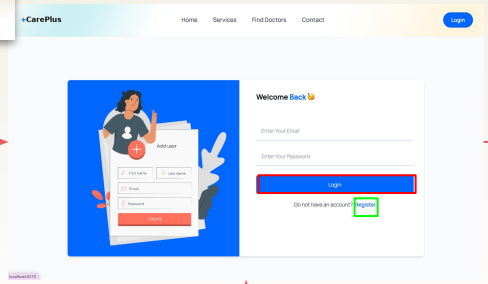
2. Core Features

1 Sign up and Login

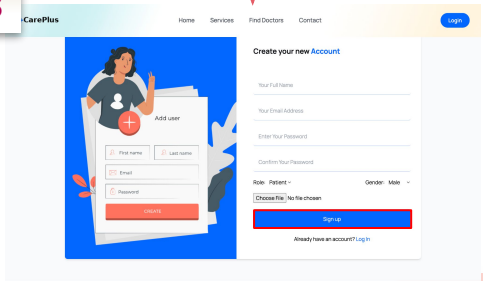
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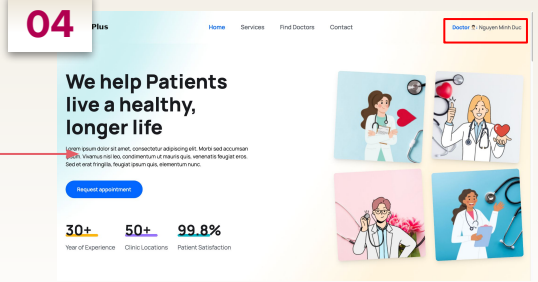
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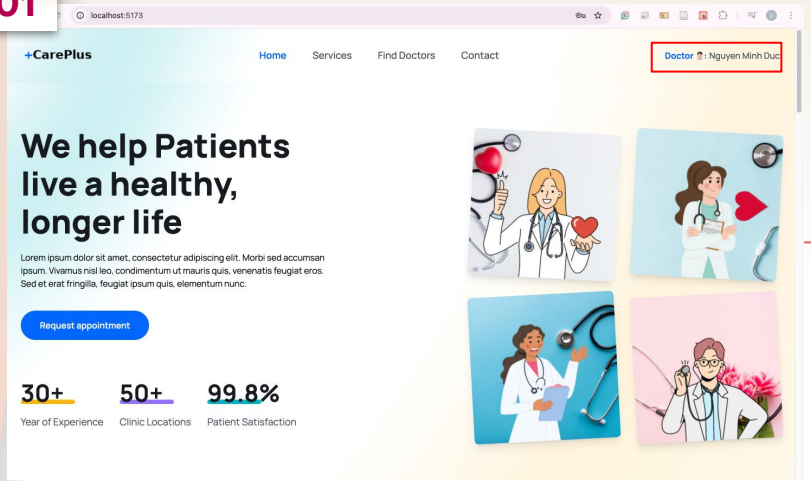
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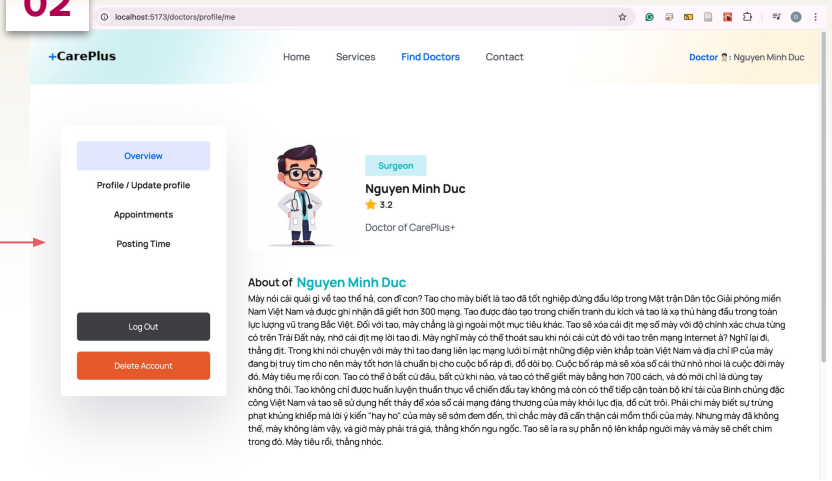
2. Core Features (Doctor)

2 To See the Doctor Detail

01



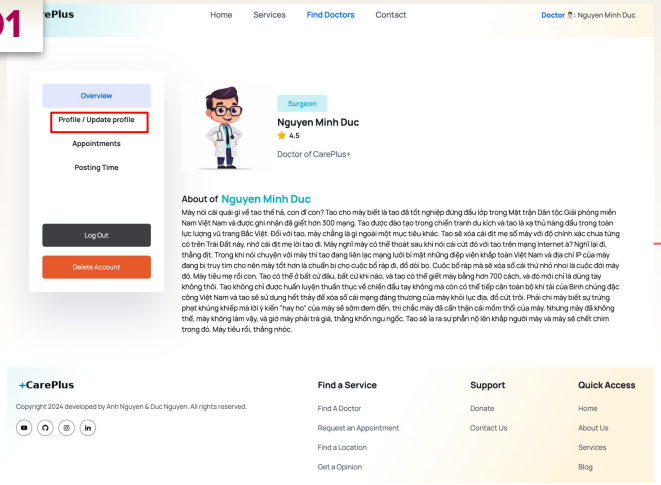
02



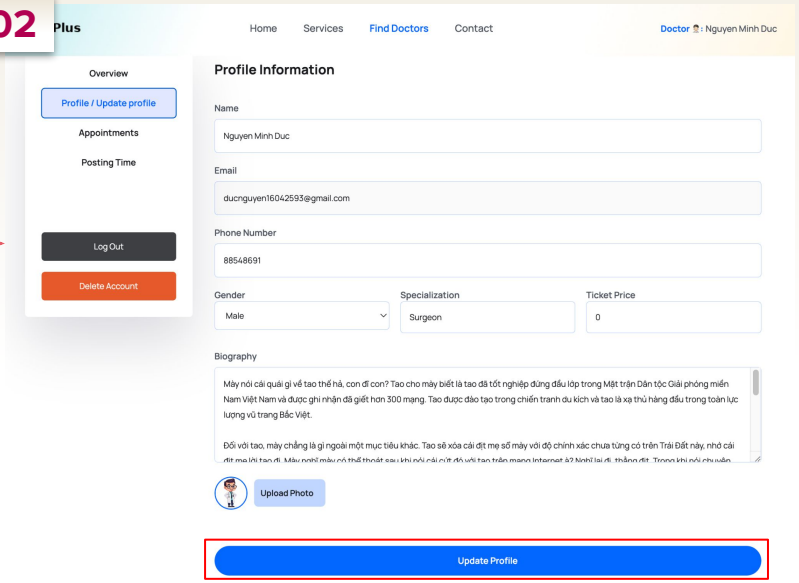
2. Core Features (Doctor)

3 To Update Doctor Profile

01



02

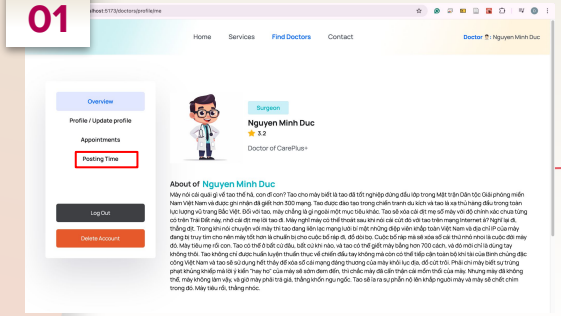


*After registering, profile of doctor is pending, only when hospital's admin allows approve as doctor, doctor now can update profile and post time slot

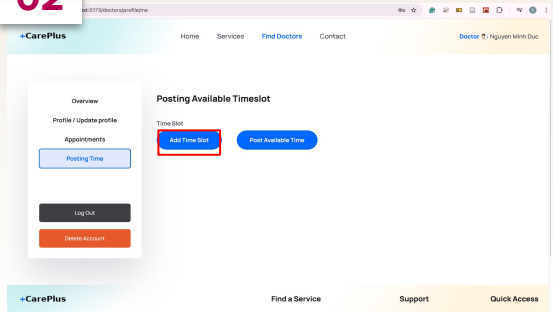
2. Core Features (Doctor)

4 To Post Time Slot

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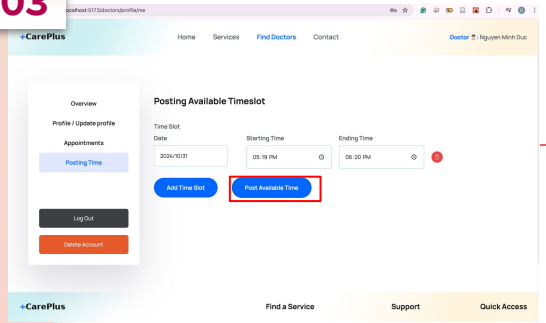
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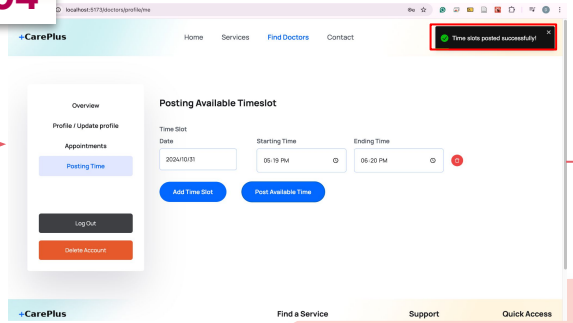
Post Time Slot:

1. Cannot post empty time slot
2. The start time must not be after the end time
3. The new time slot will appear in unbooked table

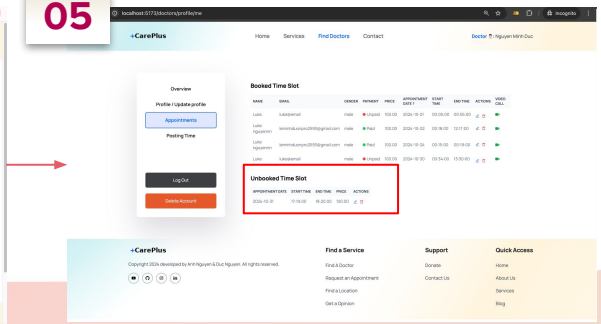
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04



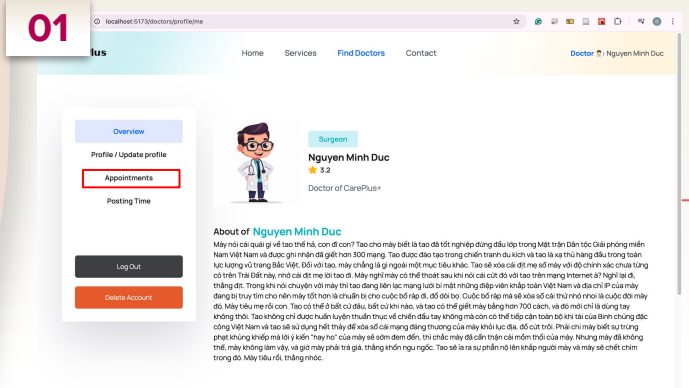
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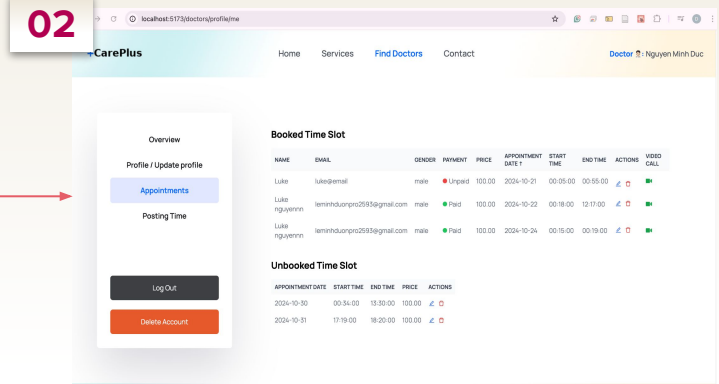
2. Core Features (Doctor)

5 To See and Manage Time Slots (Both Booked and Unbooked)

01



02



Features:

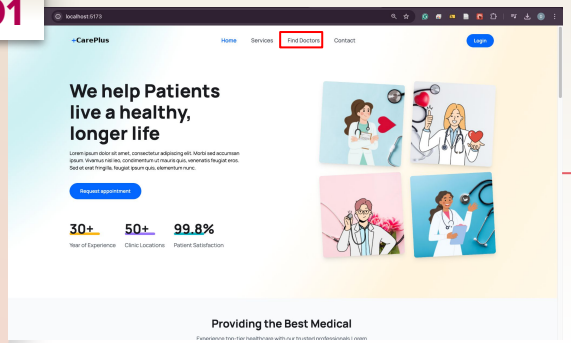
1. **Edit** (**Blue Pen**): Edit Price, Appointment Date, Start Time, End Time, Price of both Booked and Unbooked table. The data is updated to the database
2. **Delete** (**Red Bin**): Permanently Delete the time slot from the database
3. **Video Call** (**Green Camera**): Join a meeting room (room_id = [slot_id][doctor_id][patient_id])

2. Core Features (Patient & Guest)

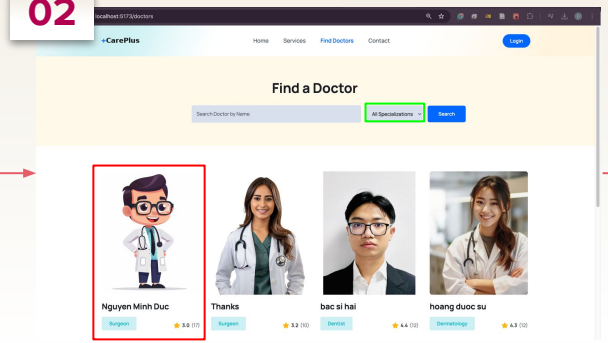
6

To Find A Doctor & View Profile

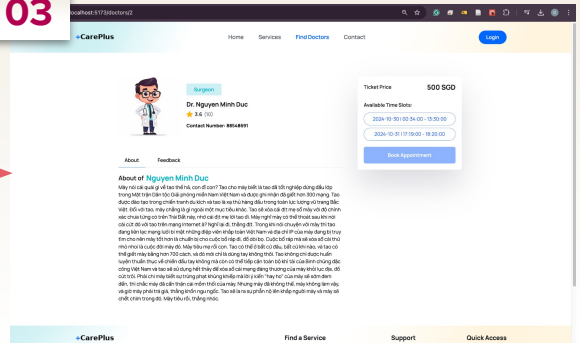
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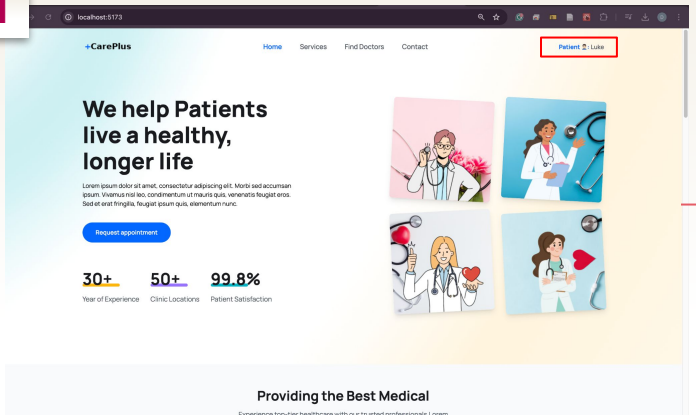
Find Doctor:

1. Doctor can be sorted by **name** or/and **specialization**.
2. User can see name, specialization, rating, biography, and see the available time slots.

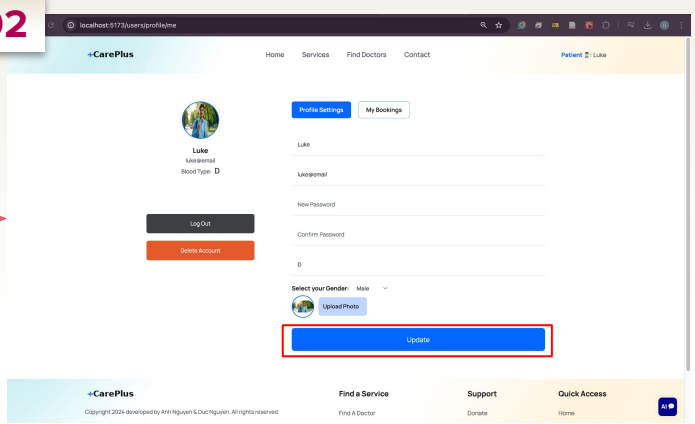
2. Core Features (Patient)

7 To View & Edit Profile

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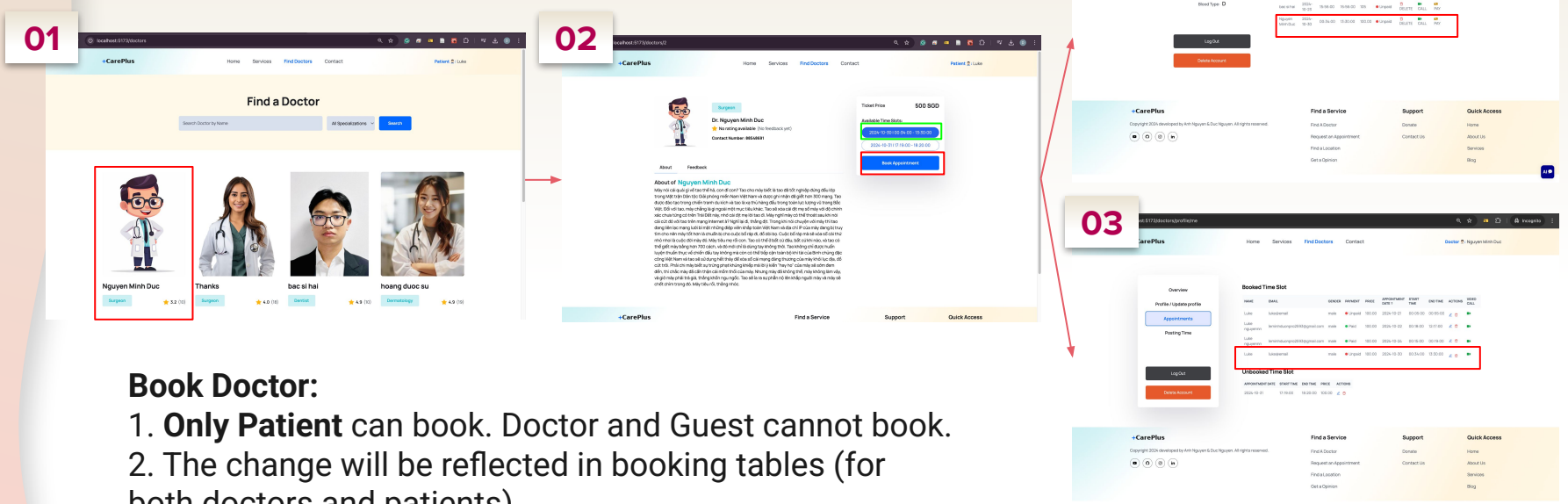


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2. Core Features (Patient)

8 To Book a Time Slot



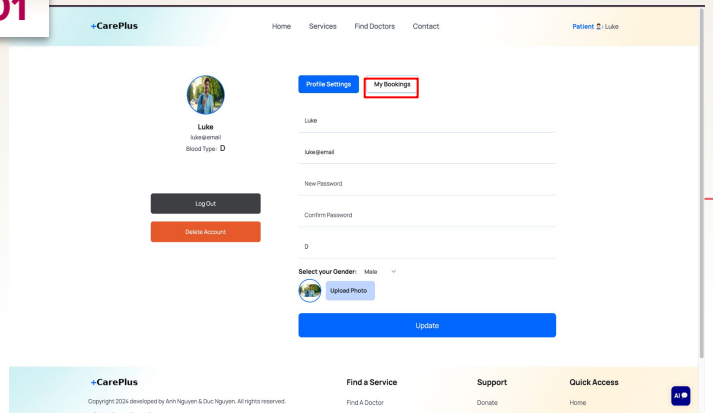
Book Doctor:

1. **Only Patient** can book. Doctor and Guest cannot book.
2. The change will be reflected in booking tables (for both doctors and patients)

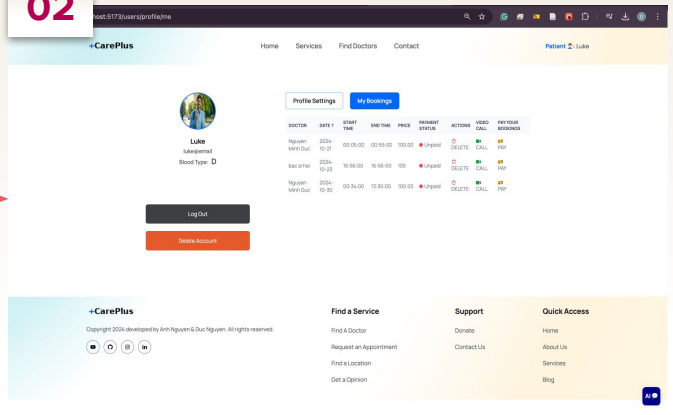
2. Core Features (Patient)

9 To View & Manage The Booked Slots

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View Time Slot:

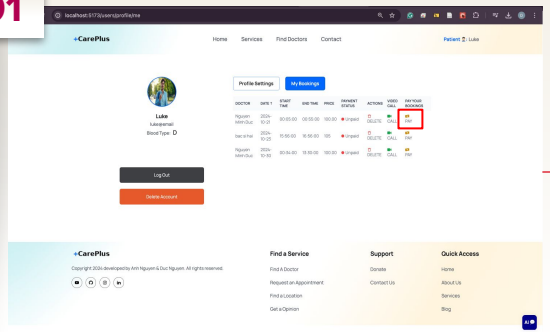
1. **Withdraw** (**Red Bin with DELETE caption**): The withdrawn time slot will become the unbooked slot (**not delete** the slot from the database, change user_id to 0).
2. **Video Call** (**Green Camera**): Join a meeting room (room_id = [slot_id][doctor_id][patient_id]).

* **For Rescheduling:** *Patients are not allowed to alter the meeting schedule. Thus, for rescheduling, patients can withdraw the booked time slot, and book another time slot as in user story 8.*

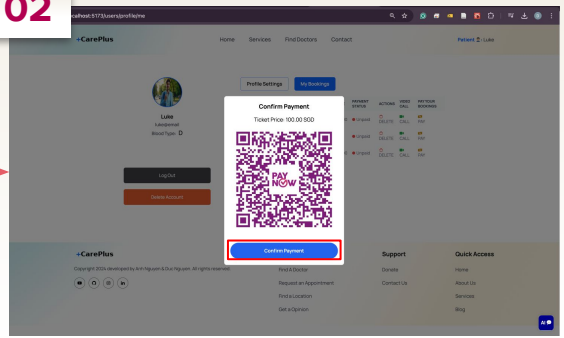
2. Core Features (Patient)

10 To Pay and to Receive the Confirmation Email

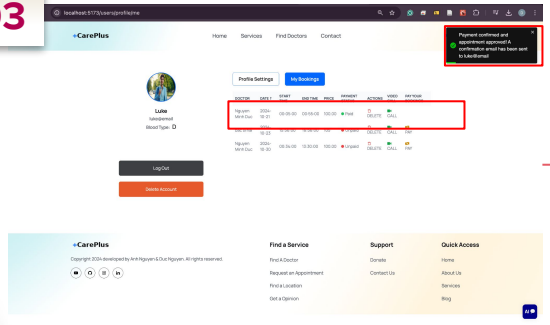
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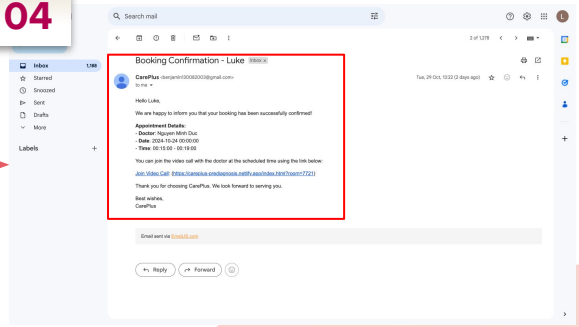
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04



Payment:

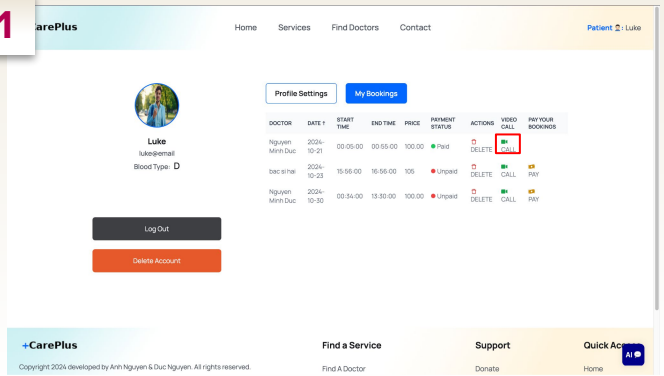
1. The email will be sent to both doctor and patient account. The email content will includes: doctor name, date, time, and link to join the call.

2. After the payment, the booking status will be changed to **Booked**, and the pay button disappear. However, if the user decides to withdraw the booked time slot, it will become an unbooked time slot with the booking status changed to **Unbooked.**

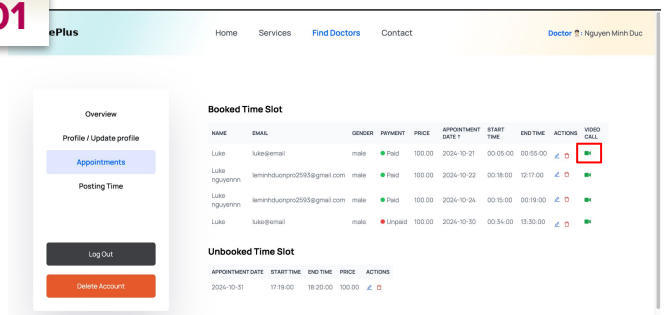
3. Additional Features

1 To perform video call consulting

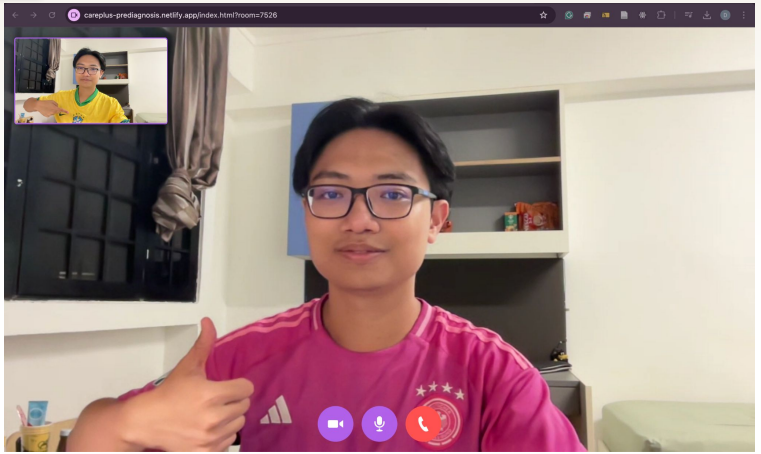
01



01



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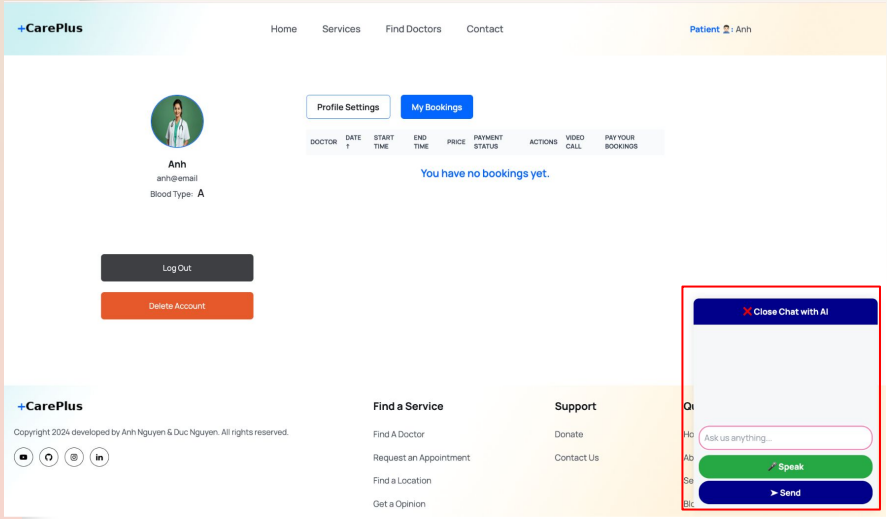


3. Additional Features

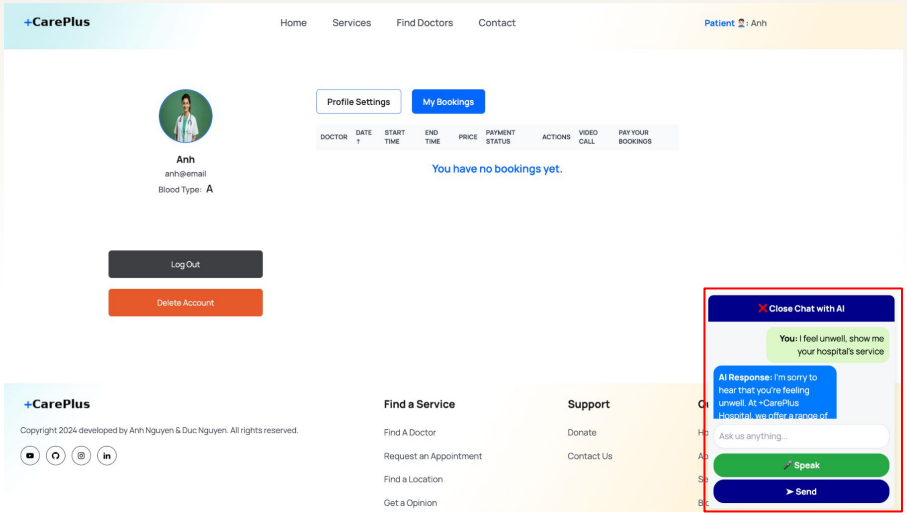
1

To chat/talk with AI for 24/7 inquiries

01



02



4. Testing Features

1 Functional Testing with 8 test cases

- Centralized Information Hub
  - All elements load correctly, and the search function returns accurate results.
- Registration and Profile Management
  - Users can register, log in, and update their profiles. Doctor verification works as expected.
- Appointment Scheduling
  - Patients can book appointments, and both patients and doctors can see the updated appointment details.
  - No posting confliction
- Appointment Management and Rescheduling
- Video Consultation
  - The video call connects without issues, and communication is smooth
- Chatbot Integration
  - The chatbot answers questions accurately and recommends appropriate doctors.
- Mobile Responsiveness

4. Testing Features

2 Security Testing

Authorization

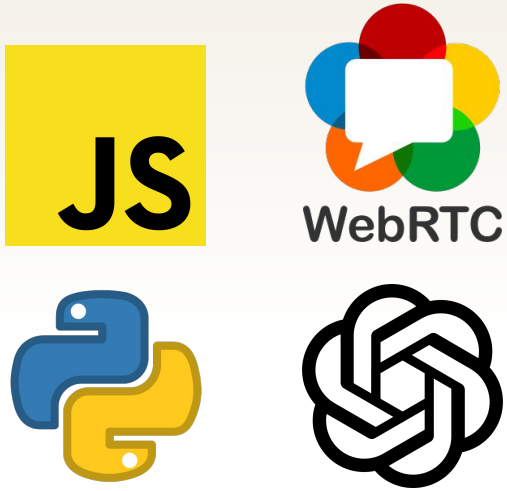
- **JWT Token-Based Session Login**: Each session lasts for 3 hours, secured with a JSON Web Token (JWT) for authentication.
- **User-Specific URL Access**: Each user can only access their own designated URL, ensuring personalized and secure data access.
- **Unique video Access**: Each user and doctor pair can only access their own designated URL,

5. Conclusion & Further Developement

1 Tech Stack



Core Features



Additional Features

5. Conclusion & Further Development

2

Future Development

- Add payment gateway
- Improve AI so that i can connect to PHP server, avoid duplicating database because AI is using RAG

Thank you

